

LLMs as few-shot learners

NLP: Fall 2025

Anoop Sarkar

"Language provides a natural domain for the study of artificial intelligence, as the vast majority of reasoning tasks can be efficiently expressed and evaluated in language, and the world's text provides a wealth of data for unsupervised learning via generative modeling."

- OpenAI

Improving Language Understanding by Generative Pre-Training

GPT1

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GPT1

Pre-training an autoregressive language model

BooksCorpus: 7K
unpublished books
(1B words)

- Start with a large amount of unlabeled data $\mathcal{U} = \{u_1, \dots, u_n\}$
- Pre-training objective: Maximize the likelihood of predicting the next token

$$L_i(\mathcal{U}) = \sum_i \log P(u_i \mid u_{i-k}, \dots, u_{i-1}; \Theta)$$

$U = (u_{-k}, \dots, u_{-1})$ is the context vector of tokens

- This is equivalent to training a Transformer decoder

n is the number of Transformer layers

$$h_0 = U \boxed{W_e} + W_p$$

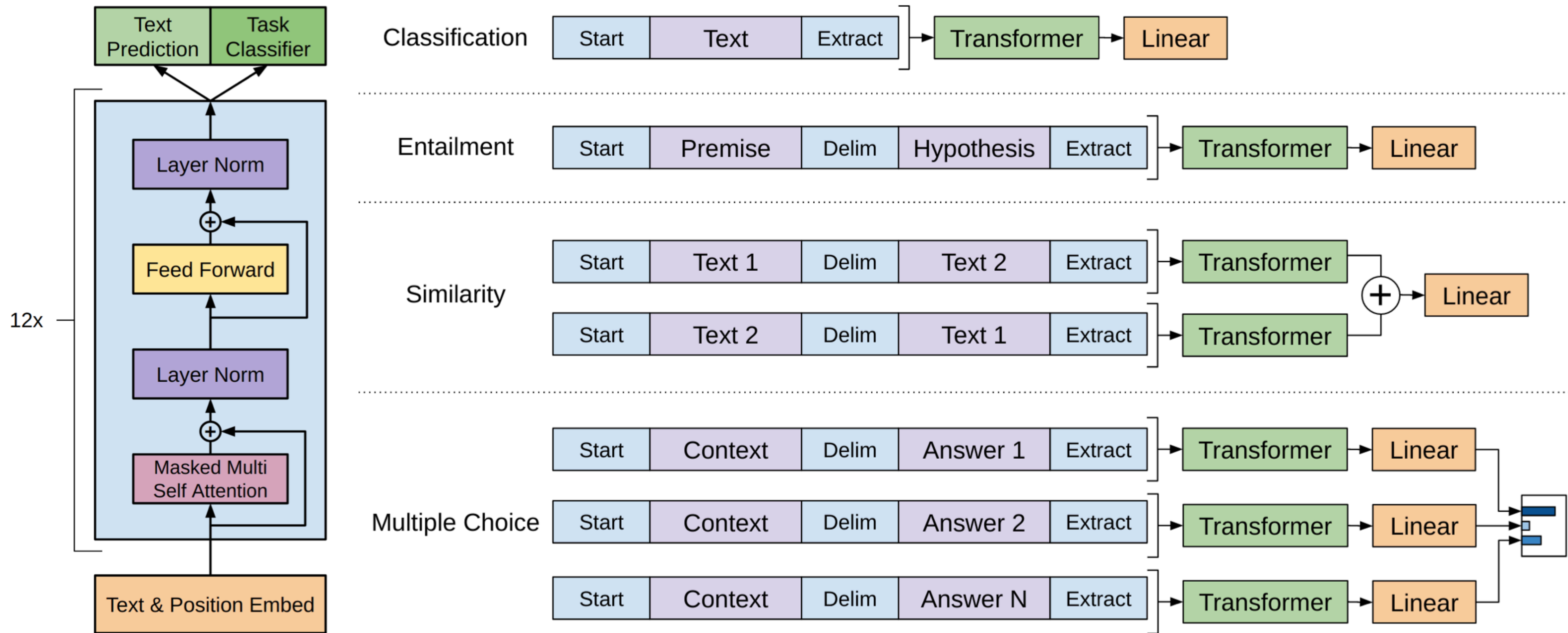
W_e is the token embedding matrix

$$h_\ell = \text{transformer_block}(h_{\ell-1}) \forall \ell \in [1, n]$$

W_p is the position embedding matrix

$$P(u) = \text{softmax}(h_n \boxed{W_e^T})$$

- Directionality is needed to generate a well-formed probability distribution



This setup was for fine-tuning GPT1 but also works for in-context learning in GPT2 and GPT3.

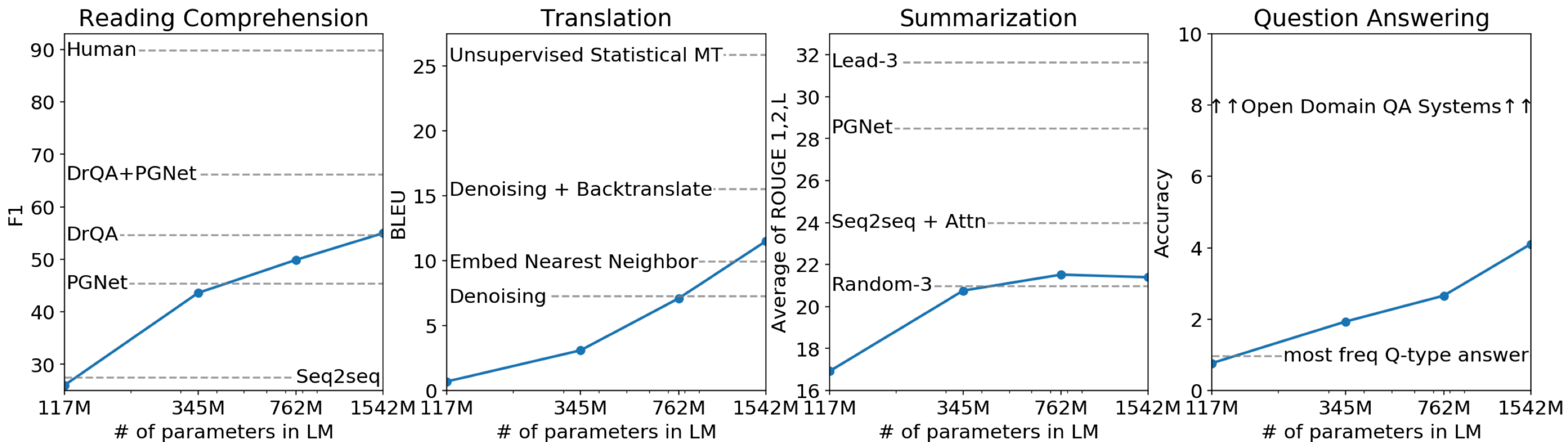
The GPT2 paper

Language Models are Unsupervised Multitask Learners

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[https://cdn.openai.com/better-language-models/
language_models_are_unsupervised_multitask_learners.pdf](https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf)

Feb 2019



WebText corpus

- Train on web scale corpus but with more reliable data compared to the CommonCrawl.
- English-only, so language detection is used
- Outgoing links from reddit (with at least 3 karma)
- No reddit data was used, instead use the content of the web sites linked on reddit discussions
- 8M documents with 40GB of text

Language detection: <https://github.com/CLD2Owners/cld2>

News site scraping: <https://github.com/codelucas/newspaper>

”I’m not the cleverest man in the world, but like they say in French: **Je ne suis pas un imbécile** [I’m not a fool].

In a now-deleted post from Aug. 16, Soheil Eid, Tory candidate in the riding of Joliette, wrote in French: **”Mentez mentez, il en restera toujours quelque chose,”** which translates as, **”Lie lie and something will always remain.”**

“I hate the word ‘**perfume**,’” Burr says. ‘It’s somewhat better in French: ‘**parfum**.’

If listened carefully at 29:55, a conversation can be heard between two guys in French: **“-Comment on fait pour aller de l’autre côté? -Quel autre côté?”**, which means **“- How do you get to the other side? - What side?”**.

If this sounds like a bit of a stretch, consider this question in French: **As-tu aller au cinéma?**, or **Did you go to the movies?**, which literally translates as Have-you to go to movies/theater?

“Brevet Sans Garantie Du Gouvernement”, translated to English: **“Patented without government warranty”**.

Table 1. Examples of naturally occurring demonstrations of English to French and French to English translation found throughout the WebText training set.

Parameters	Layers	d_{model}
117M	12	768
345M	24	1024
762M	36	1280
1542M	48	1600

Table 2. Architecture hyperparameters for the 4 model sizes.

Perplexity Results

	LAMBADA (PPL)	LAMBADA (ACC)	CBT-CN (ACC)	CBT-NE (ACC)	WikiText2 (PPL)	PTB (PPL)	enwik8 (BPB)	text8 (BPC)	WikiText103 (PPL)	1BW (PPL)
SOTA	99.8	59.23	85.7	82.3	39.14	46.54	0.99	1.08	18.3	21.8
117M	35.13	45.99	87.65	83.4	29.41	65.85	1.16	1.17	37.50	75.20
345M	15.60	55.48	92.35	87.1	22.76	47.33	1.01	1.06	26.37	55.72
762M	10.87	60.12	93.45	88.0	19.93	40.31	0.97	1.02	22.05	44.575
1542M	8.63	63.24	93.30	89.05	18.34	35.76	0.93	0.98	17.48	42.16

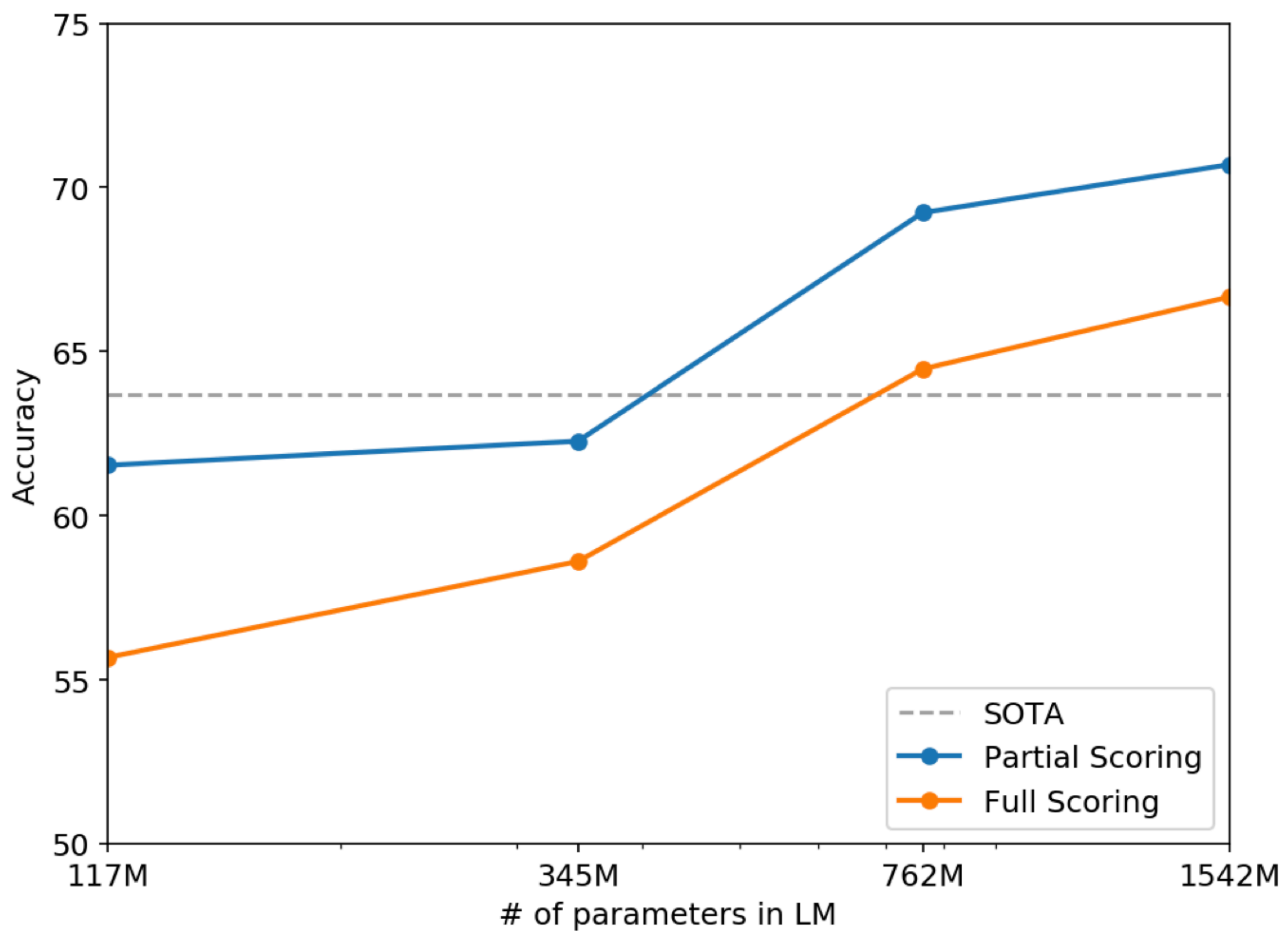


Figure 3. Performance on the Winograd Schema Challenge as a function of model capacity.

Language Models are Few-Shot Learners

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Traditional fine-tuning (not used for GPT-3)

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



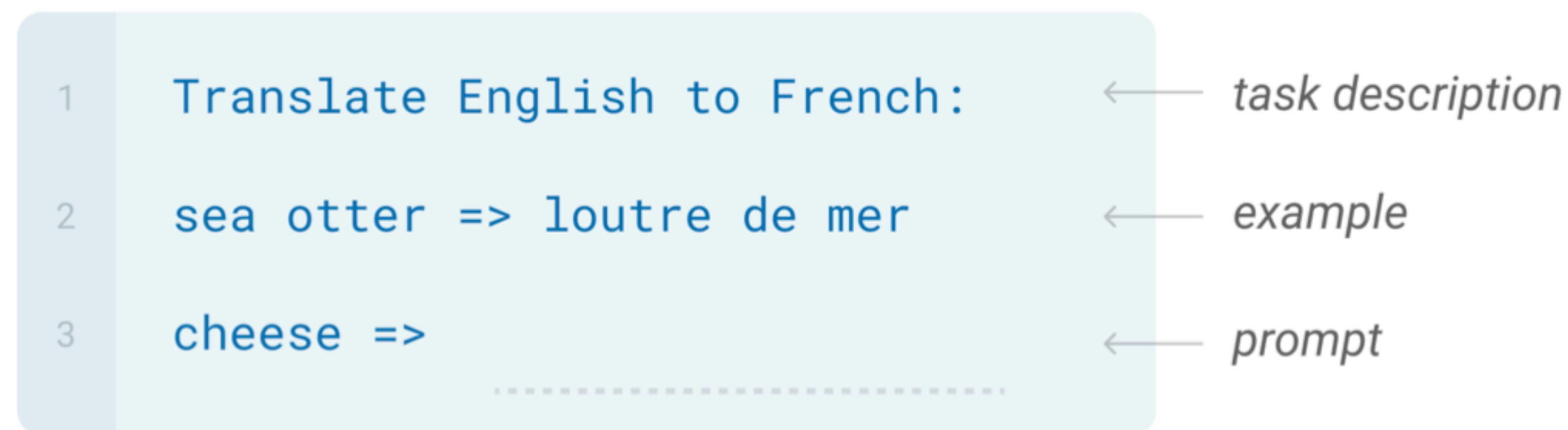
Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.



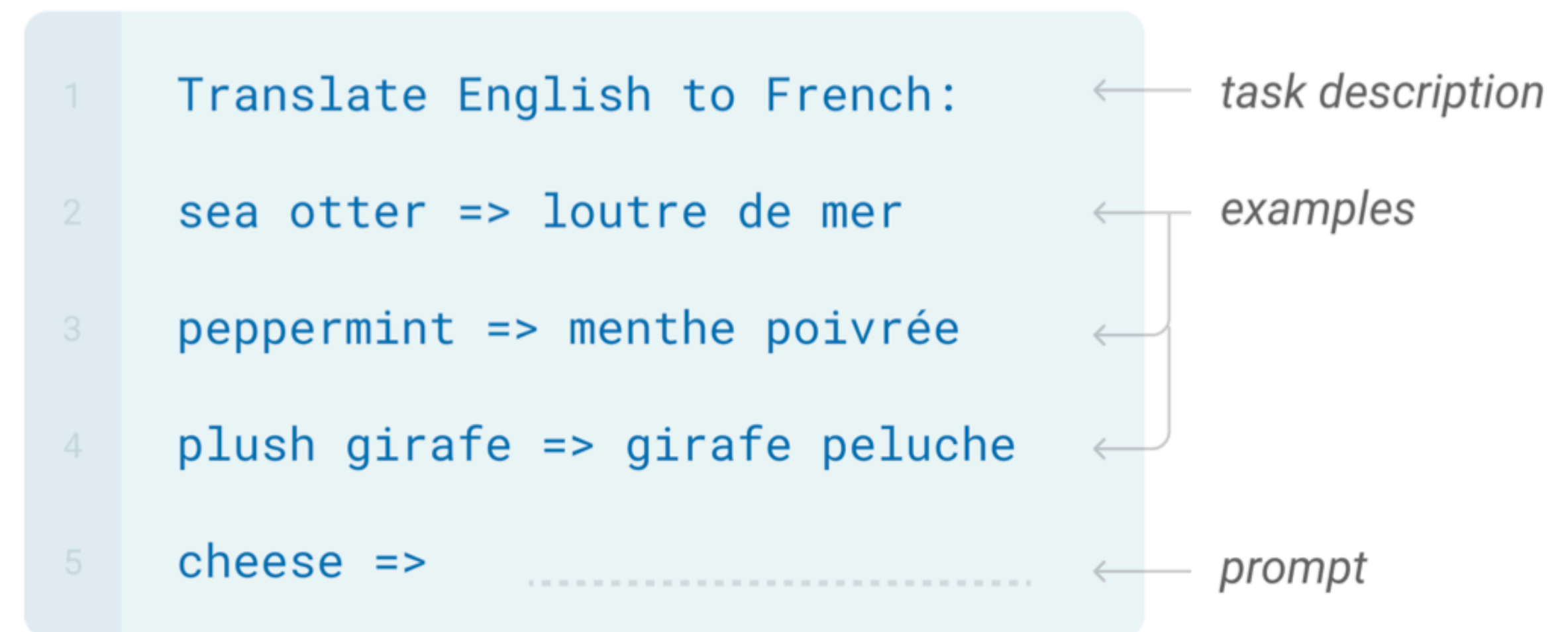
One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.



Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



Fine-tuning fails at scale

- LLMs $>10\text{B}$ parameters are very difficult to fine-tune and requires a big compute budget
- So in-context learning using a long prompt or prefix is needed to coax the answer from a "predict the next token" approach to solving multiple tasks
- Pre-training on web-scale text can observe many different tasks in-context during training in the inner loop (per batch)
- Gradient descent improves the model representations based on next token prediction over many batch updates in the outer loop

outer loop

Learning via SGD during unsupervised pre-training

inner loop

In-context learning

sequence #1

1	$5 + 8 = 13$
2	$7 + 2 = 9$
3	$1 + 0 = 1$
4	$3 + 4 = 7$
5	$5 + 9 = 14$
6	$9 + 8 = 17$

In-context learning

sequence #2

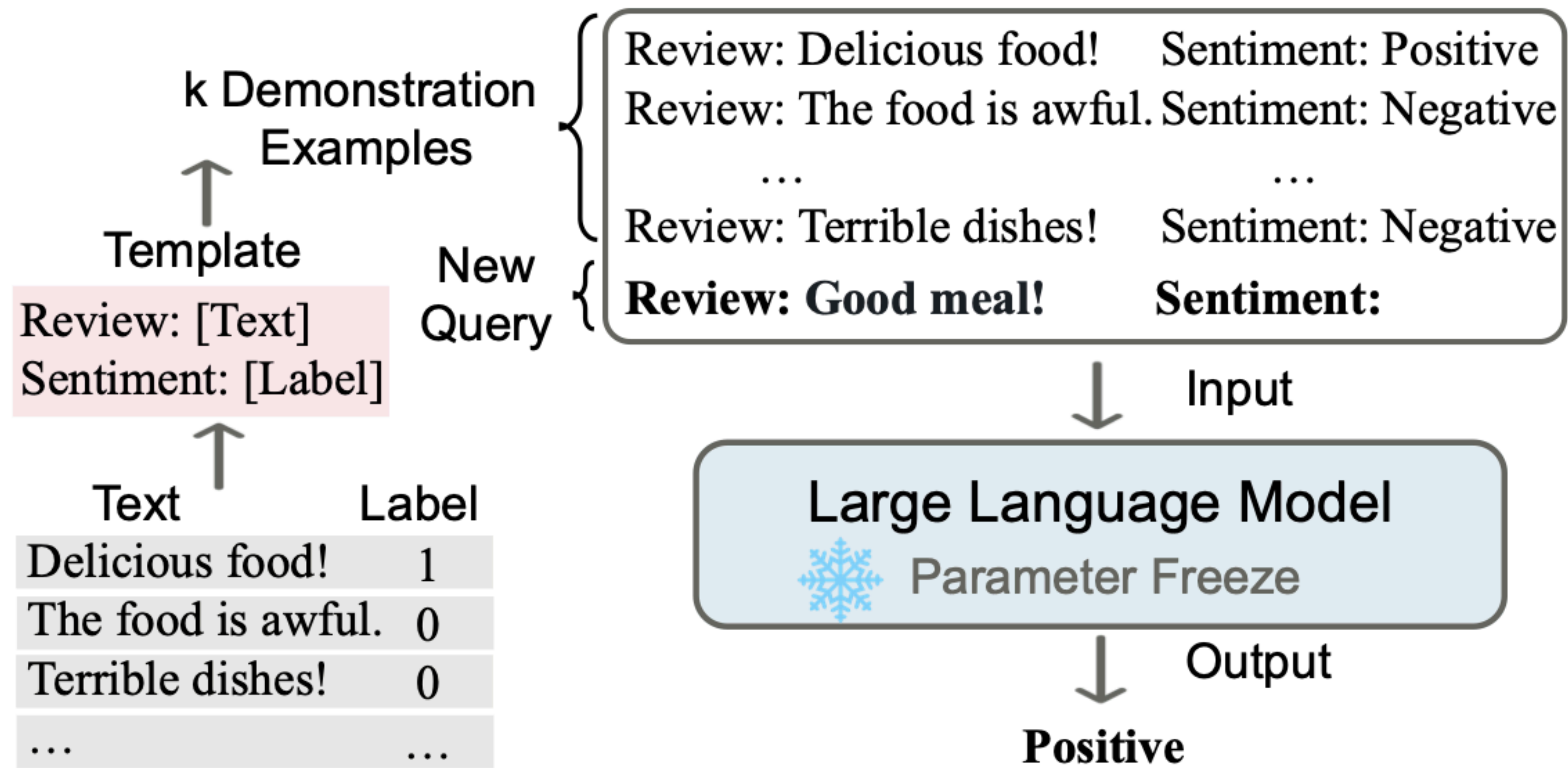
1	gaot => goat
2	sakne => snake
3	brid => bird
4	fsih => fish
5	dcuk => duck
6	cmihp => chimp

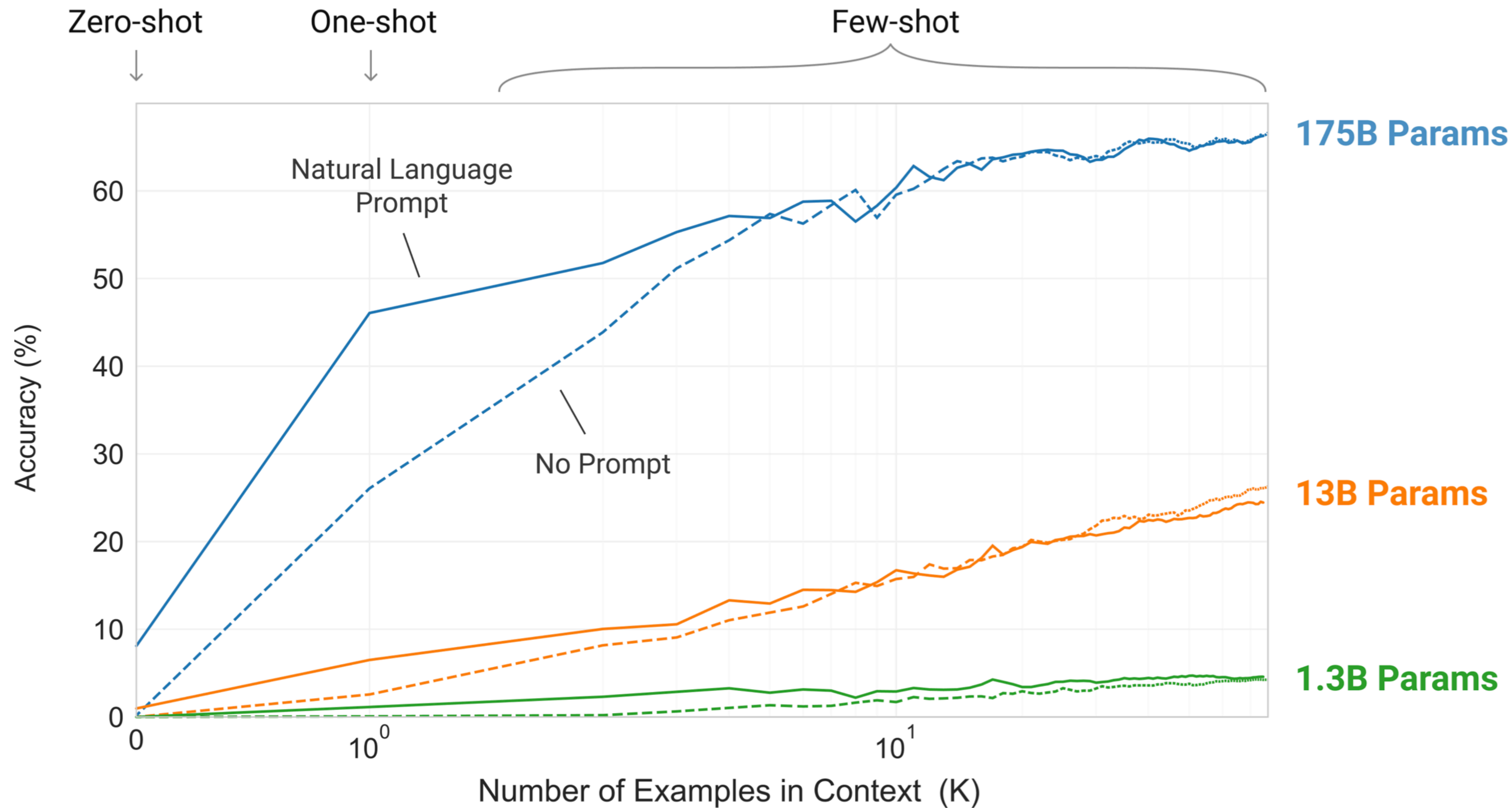
In-context learning

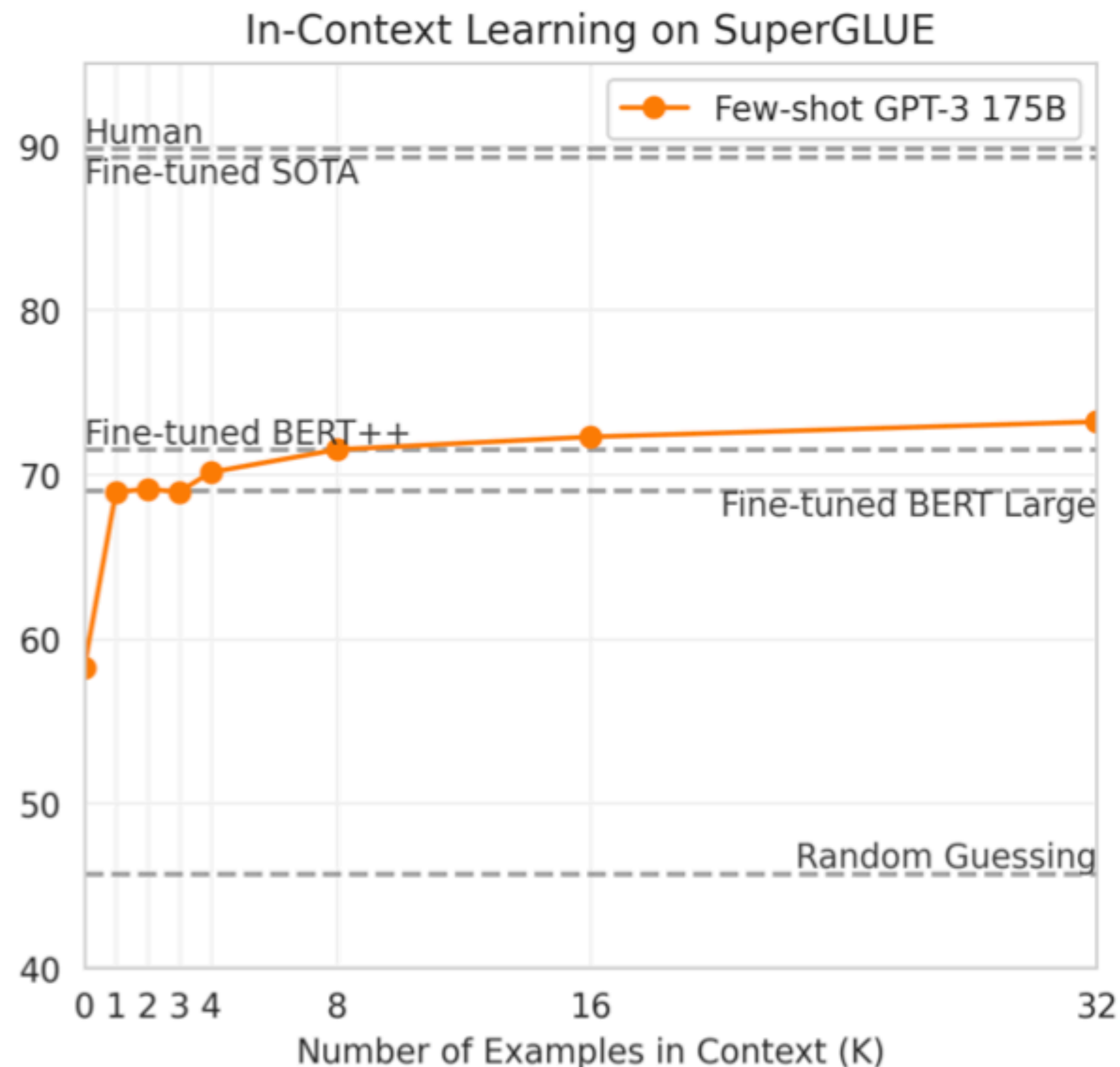
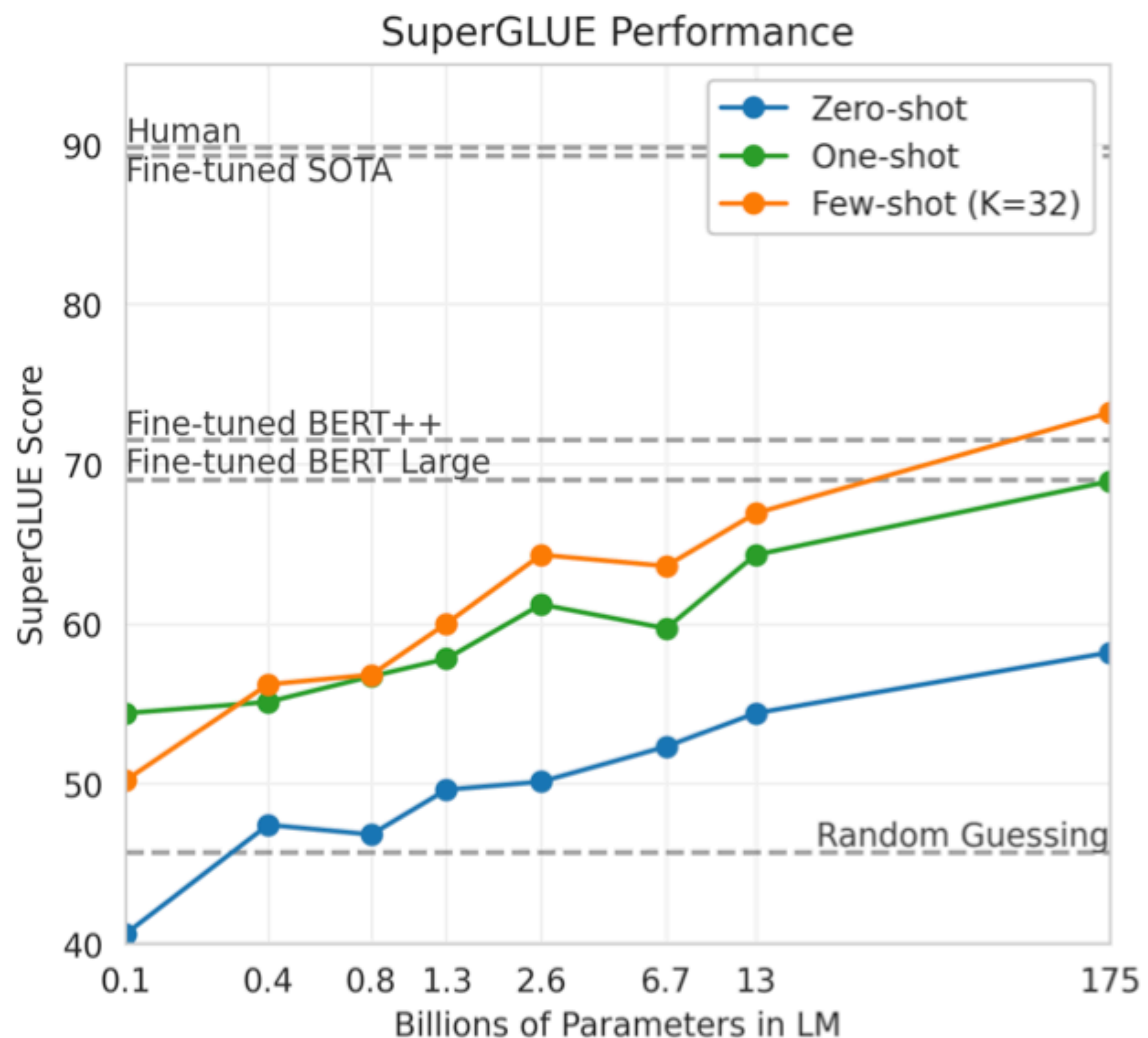
sequence #3

1	thanks => merci
2	hello => bonjour
3	mint => menthe
4	wall => mur
5	otter => loutre
6	bread => pain

Inference time







Performance on SuperGLUE increases with number of examples in context. We find the difference in performance between the BERT-Large and BERT++ to be roughly equivalent to the difference between GPT-3 with one example per context versus eight examples per context.

	SuperGLUE Average	BoolQ Accuracy	CB Accuracy	CB F1	COPA Accuracy	RTE Accuracy
Fine-tuned SOTA	89.0	91.0	96.9	93.9	94.8	92.5
Fine-tuned BERT-Large	69.0	77.4	83.6	75.7	70.6	71.7
GPT-3 Few-Shot	71.8	76.4	75.6	52.0	92.0	69.0
	WiC Accuracy	WSC Accuracy	MultiRC Accuracy	MultiRC F1a	ReCoRD Accuracy	ReCoRD F1
Fine-tuned SOTA	76.1	93.8	62.3	88.2	92.5	93.3
Fine-tuned BERT-Large	69.6	64.6	24.1	70.0	71.3	72.0
GPT-3 Few-Shot	49.4	80.1	30.5	75.4	90.2	91.1

Table 3.5: Performance of GPT-3 on SuperGLUE compared to fine-tuned baselines and SOTA. All results are reported on the test set. GPT-3 few-shot is given a total of 32 examples within the context of each task and performs no gradient updates.

Setting	LAMBADA (acc)	LAMBADA (ppl)	StoryCloze (acc)	HellaSwag (acc)
SOTA	68.0 ^a	8.63 ^b	91.8^c	85.6^d
GPT-3 Zero-Shot	76.2	3.00	83.2	78.9
GPT-3 One-Shot	72.5	3.35	84.7	78.1
GPT-3 Few-Shot	86.4	1.92	87.7	79.3

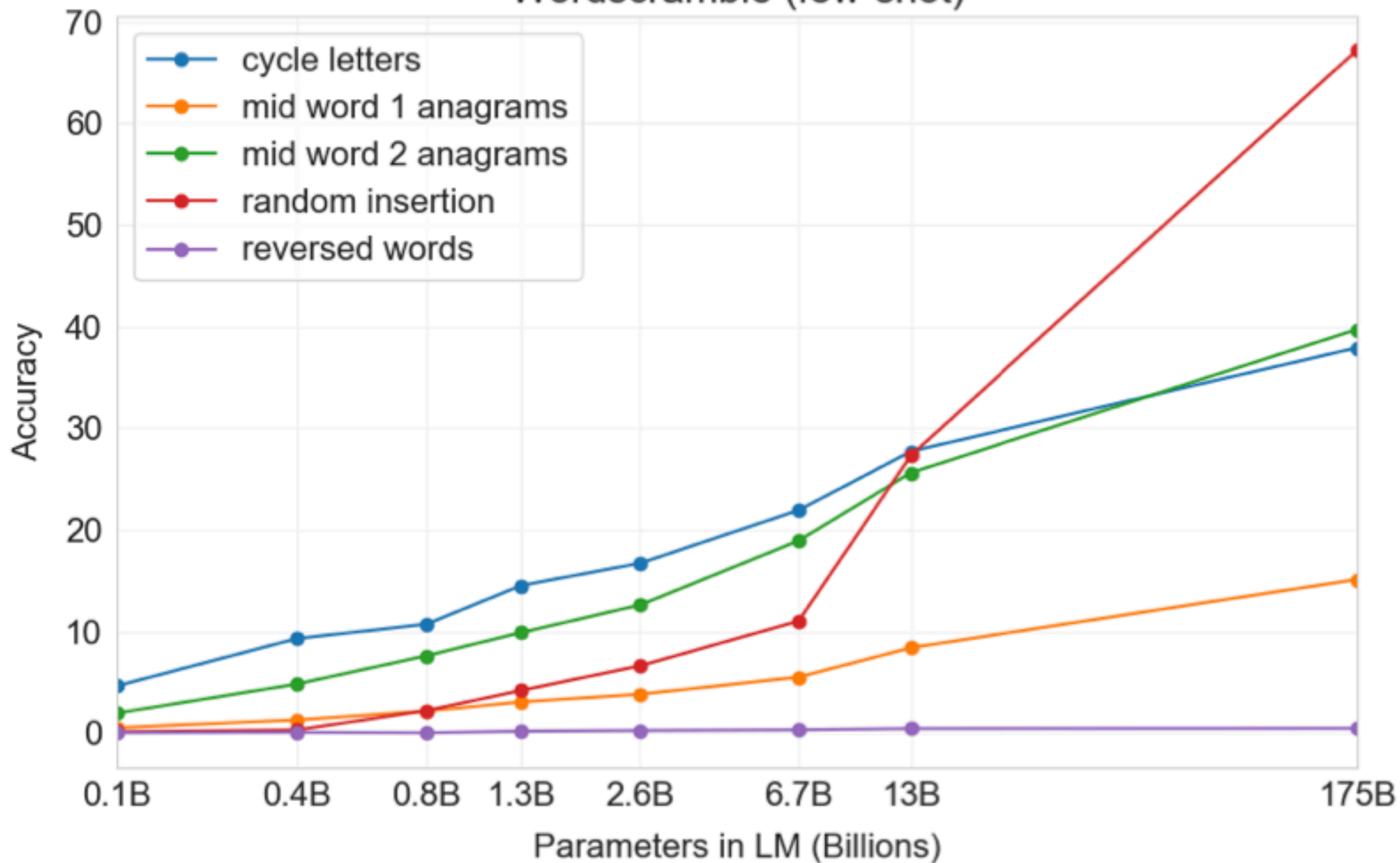
Setting	NaturalQS	WebQS	TriviaQA
RAG (Fine-tuned, Open-Domain) [LPP⁺20]	44.5	45.5	68.0
T5-11B+SSM (Fine-tuned, Closed-Book) [RRS20]	36.6	44.7	60.5
T5-11B (Fine-tuned, Closed-Book)	34.5	37.4	50.1
GPT-3 Zero-Shot	14.6	14.4	64.3
GPT-3 One-Shot	23.0	25.3	68.0
GPT-3 Few-Shot	29.9	41.5	71.2

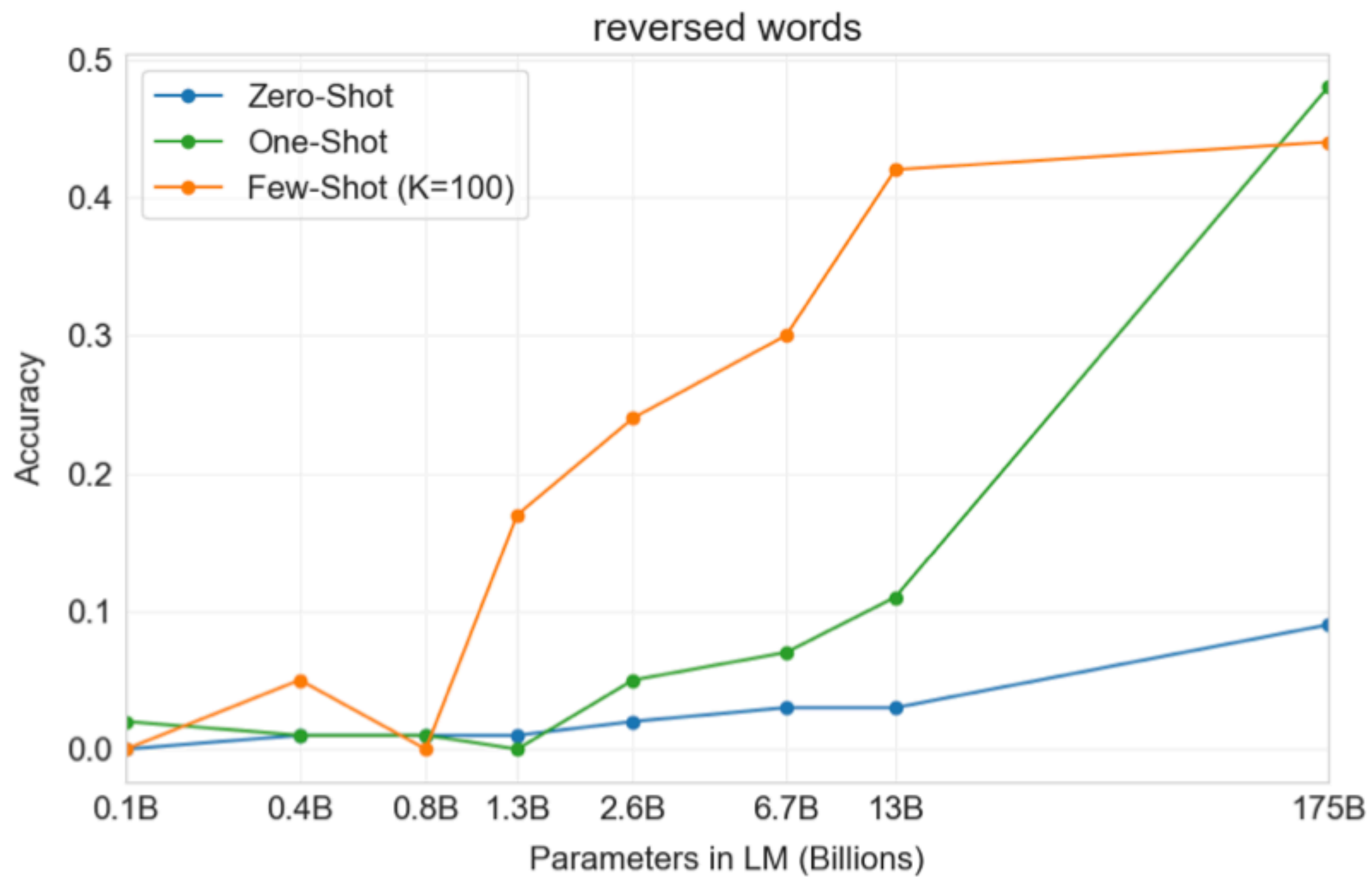
Setting	ARC (Easy)	ARC (Challenge)	CoQA	DROP
Fine-tuned SOTA	92.0^a	78.5^b	90.7^c	89.1^d
GPT-3 Zero-Shot	68.8	51.4	81.5	23.6
GPT-3 One-Shot	71.2	53.2	84.0	34.3
GPT-3 Few-Shot	70.1	51.5	85.0	36.5

WMT 2014

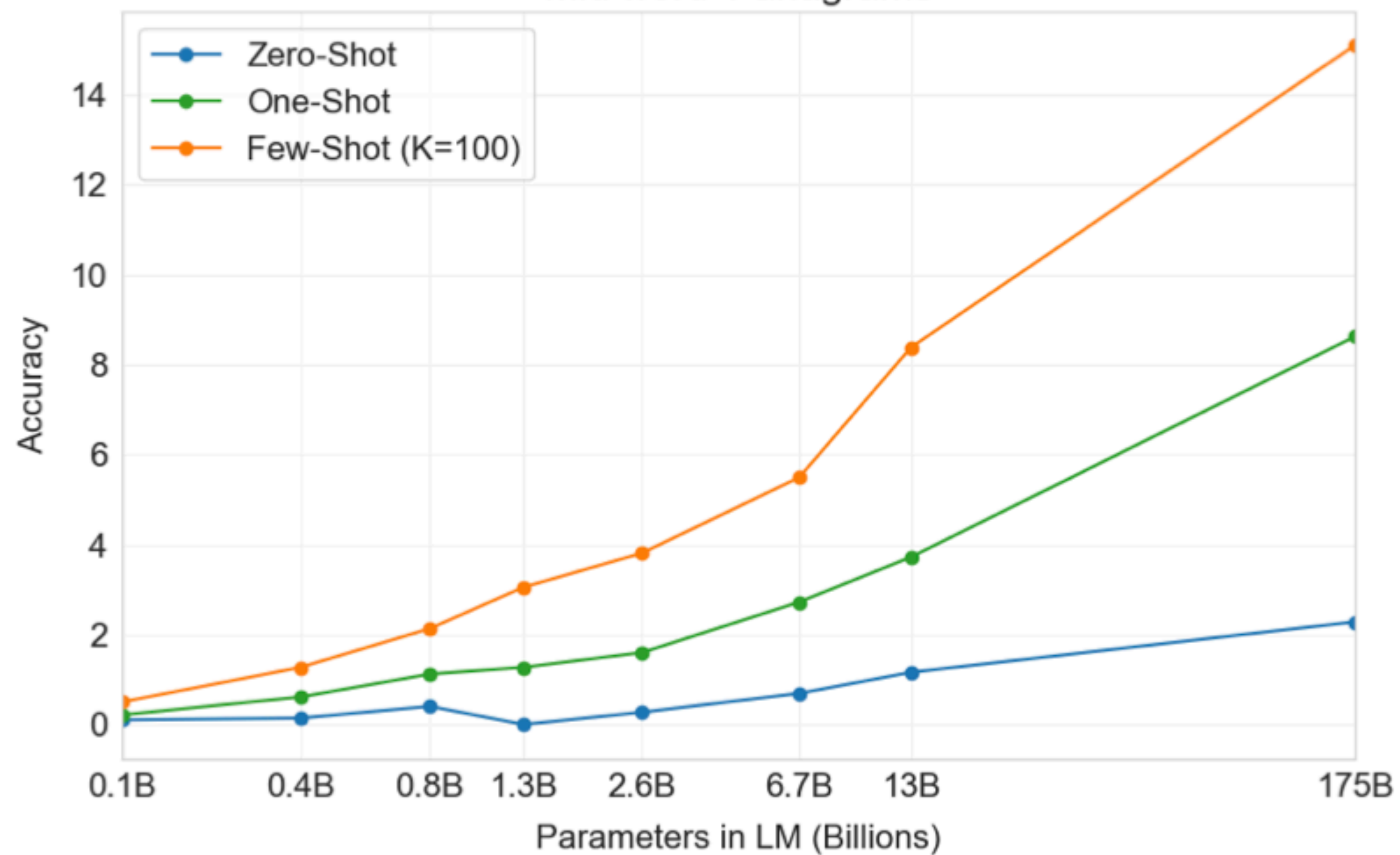
Setting	En→Fr	Fr→En	En→De	De→En	En→Ro	Ro→En
SOTA (Supervised)	45.6^a	35.0 ^b	41.2^c	40.2 ^d	38.5^e	39.9^e
XLM [LC19]	33.4	33.3	26.4	34.3	33.3	31.8
MASS [STQ ⁺ 19]	<u>37.5</u>	34.9	28.3	35.2	<u>35.2</u>	33.1
mBART [LGG ⁺ 20]	-	-	<u>29.8</u>	34.0	35.0	30.5
GPT-3 Zero-Shot	25.2	21.2	24.6	27.2	14.1	19.9
GPT-3 One-Shot	28.3	33.7	26.2	30.4	20.6	38.6
GPT-3 Few-Shot	32.6	<u>39.2</u>	29.7	<u>40.6</u>	21.0	<u>39.5</u>

Wordscramble (few-shot)

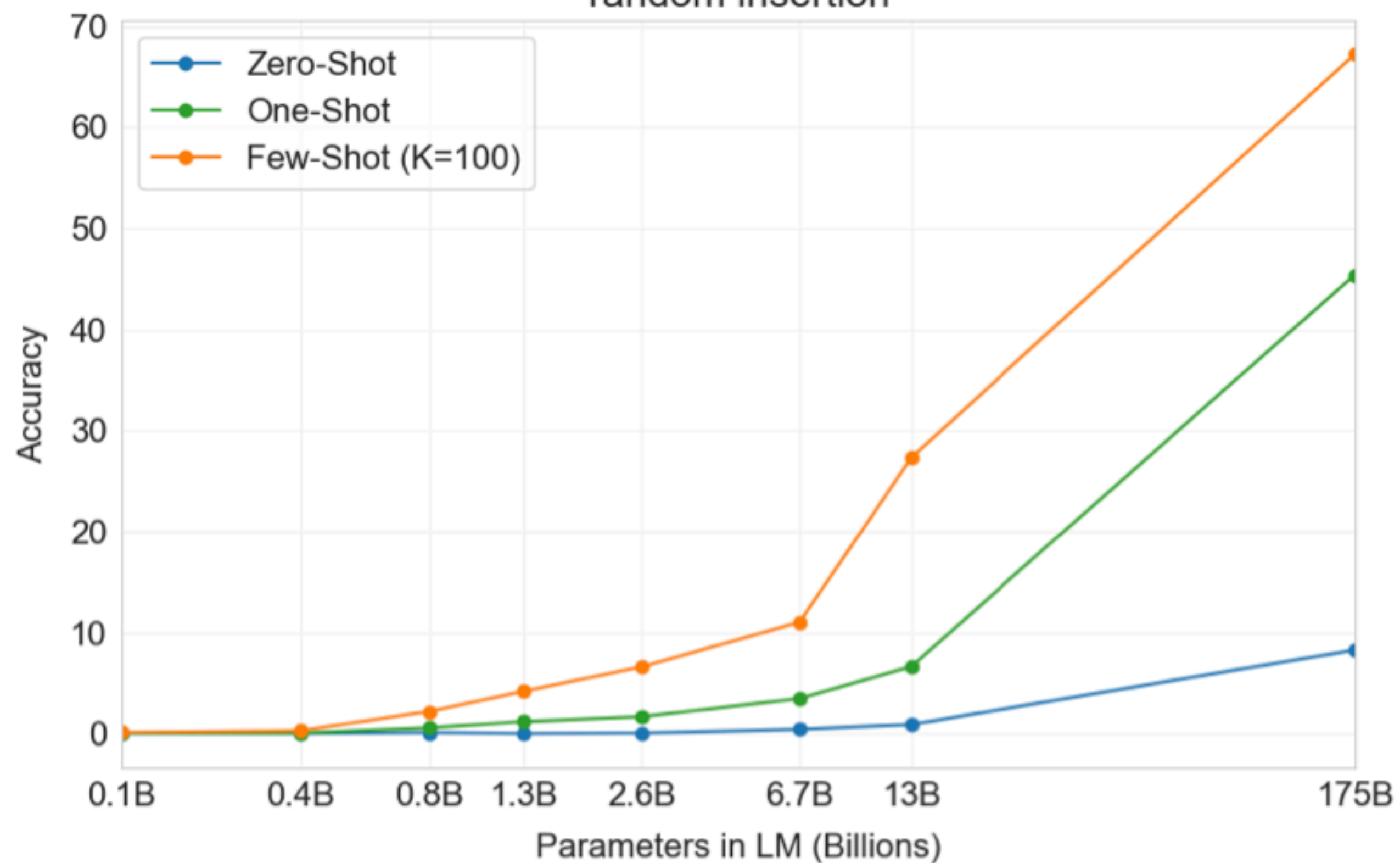




mid word 1 anagrams



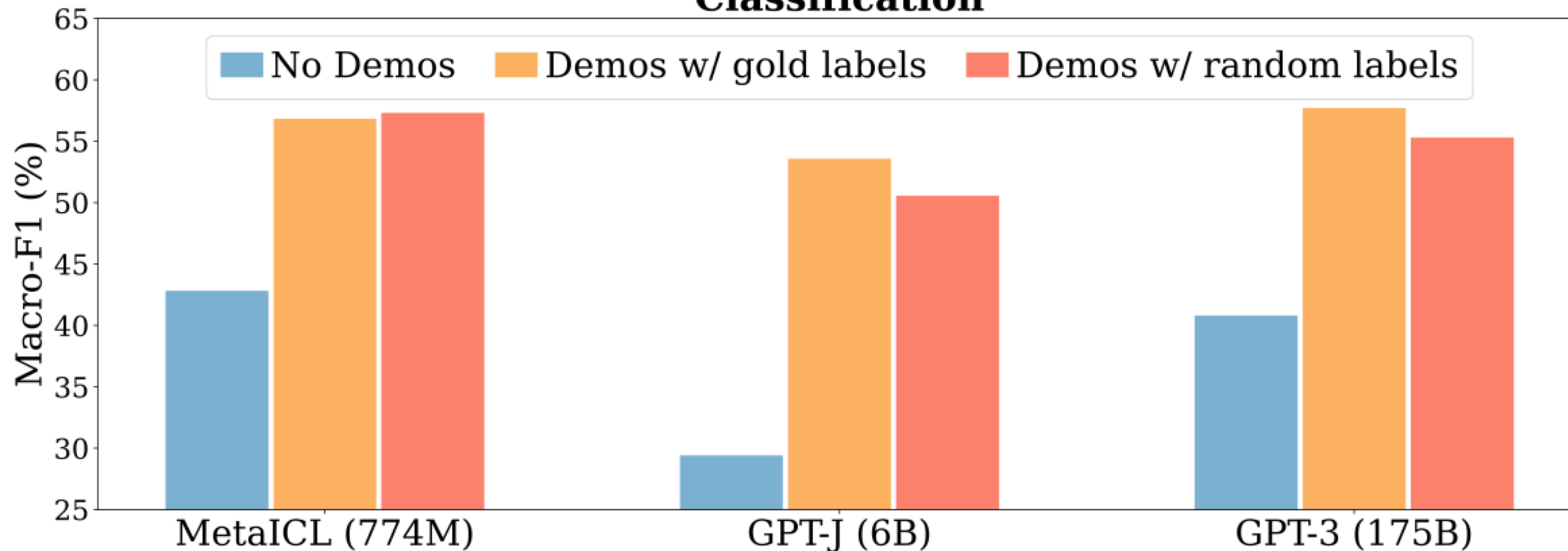
random insertion



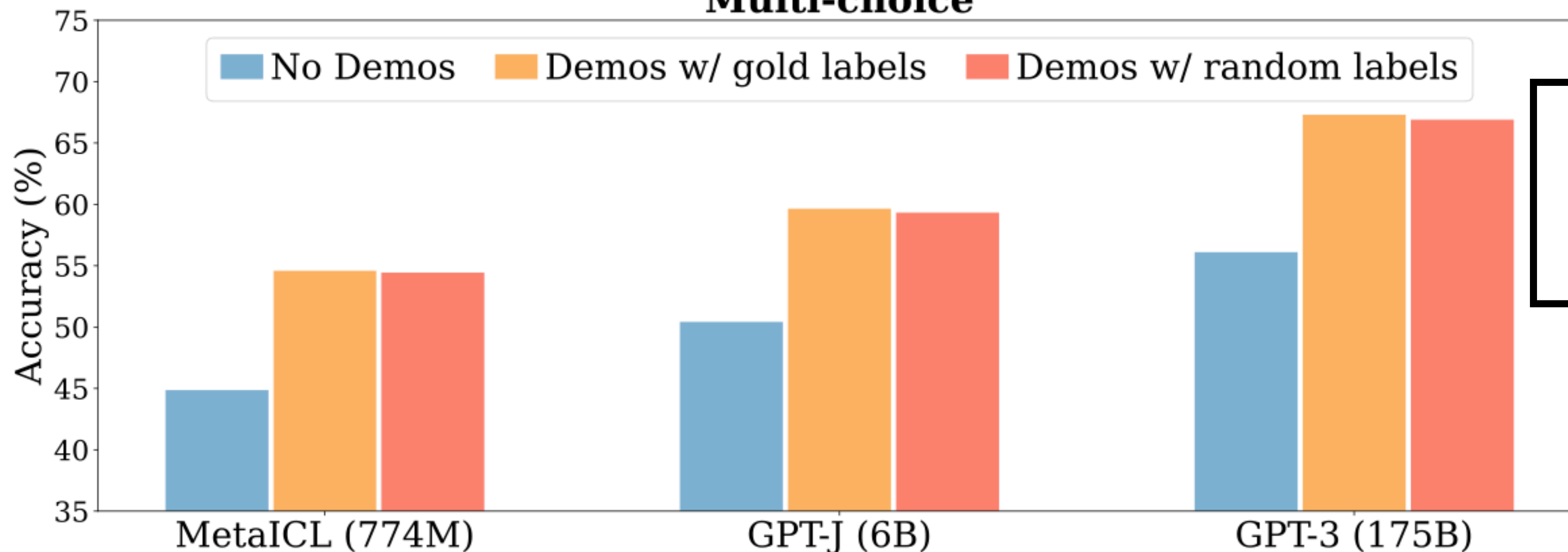
Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?

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Mike Lewis² Hannaneh Hajishirzi^{1,3} Luke Zettlemoyer^{1,2}
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Classification



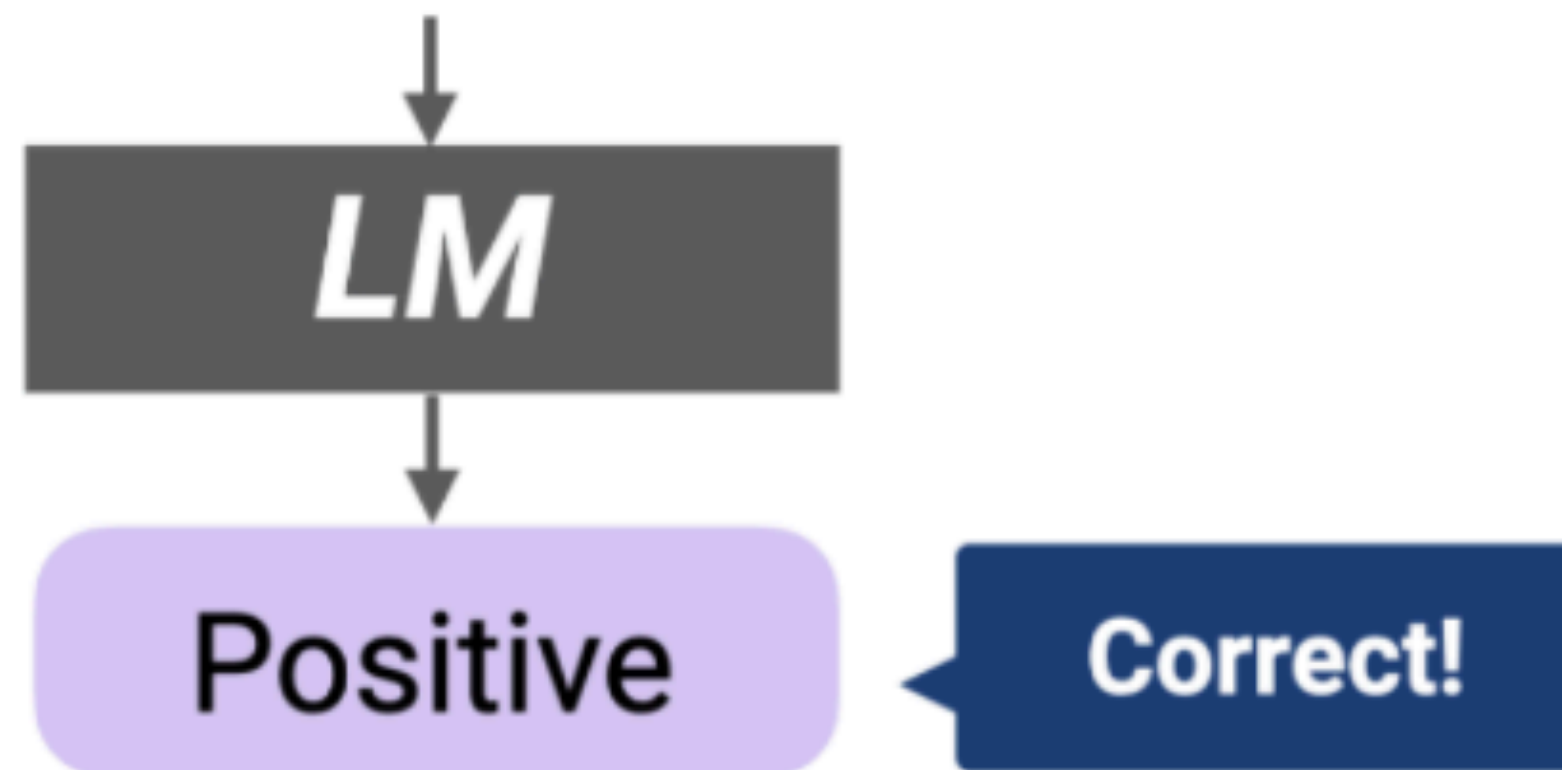
Multi-choice



ground truth
labels don't
matter!

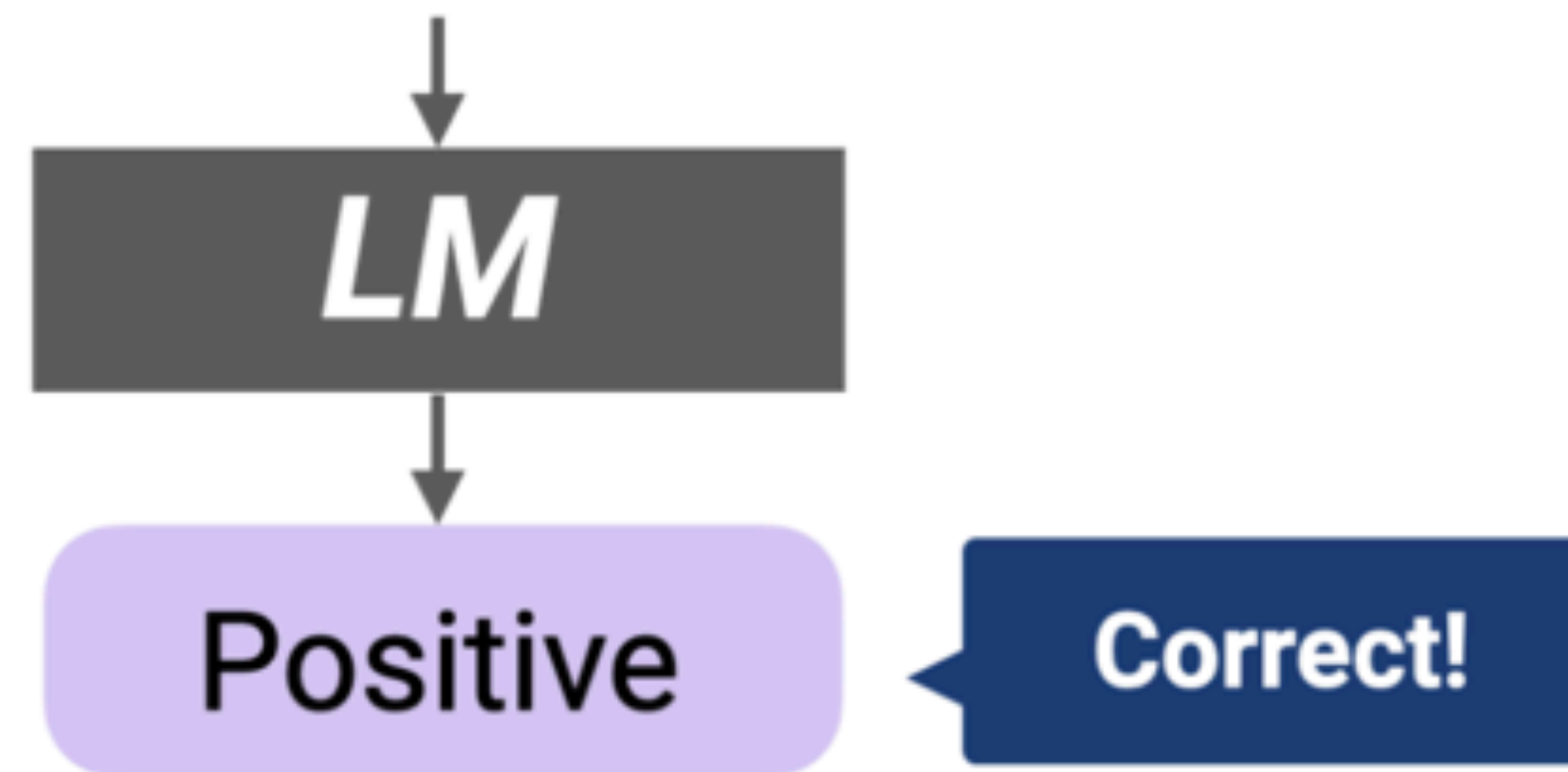
ground truth
labels

Circulation revenue has increased by 5% in Finland.	\n	Positive
Panostaja did not disclose the purchase price.	\n	Neutral
Paying off the national debt will be extremely painful.	\n	Negative
The company anticipated its operating profit to improve.	\n	_____



replace true labels with
random labels

Circulation revenue has increased by 5% in Finland. \n **Neutral**
Panostaja did not disclose the purchase price. \n **Negative**
Paying off the national debt will be extremely painful. \n **Positive**
The company anticipated its operating profit to improve. \n _____



Why does in-context learning work?

Four hypotheses

1. The input-label mapping, whether each input x_i is paired with the correct label y_i (not true)
2. The distribution that the input $x_1, \dots, x_k$ are from (is it from a sports article, or business news?)
3. The output label space $y_1, \dots, y_k$
4. The format of the demonstration, e.g. $x \quad // \quad y$; Input: x Output: y ; etc.

Demonstrations

Distribution of inputs

Label space

Circulation revenue has increased by 5% in Finland.	\n	Positive
Panostaja did not disclose the purchase price.	\n	Neutral
Paying off the national debt will be extremely painful.	\n	Negative

***Format
(The use
of pairs)***

Test example

The acquisition will have an immediate positive impact.	\n	?
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Input-label mapping

Colour-printed lithograph. Very good condition.

\n Neutral

Many accompanying marketing ... meaning.

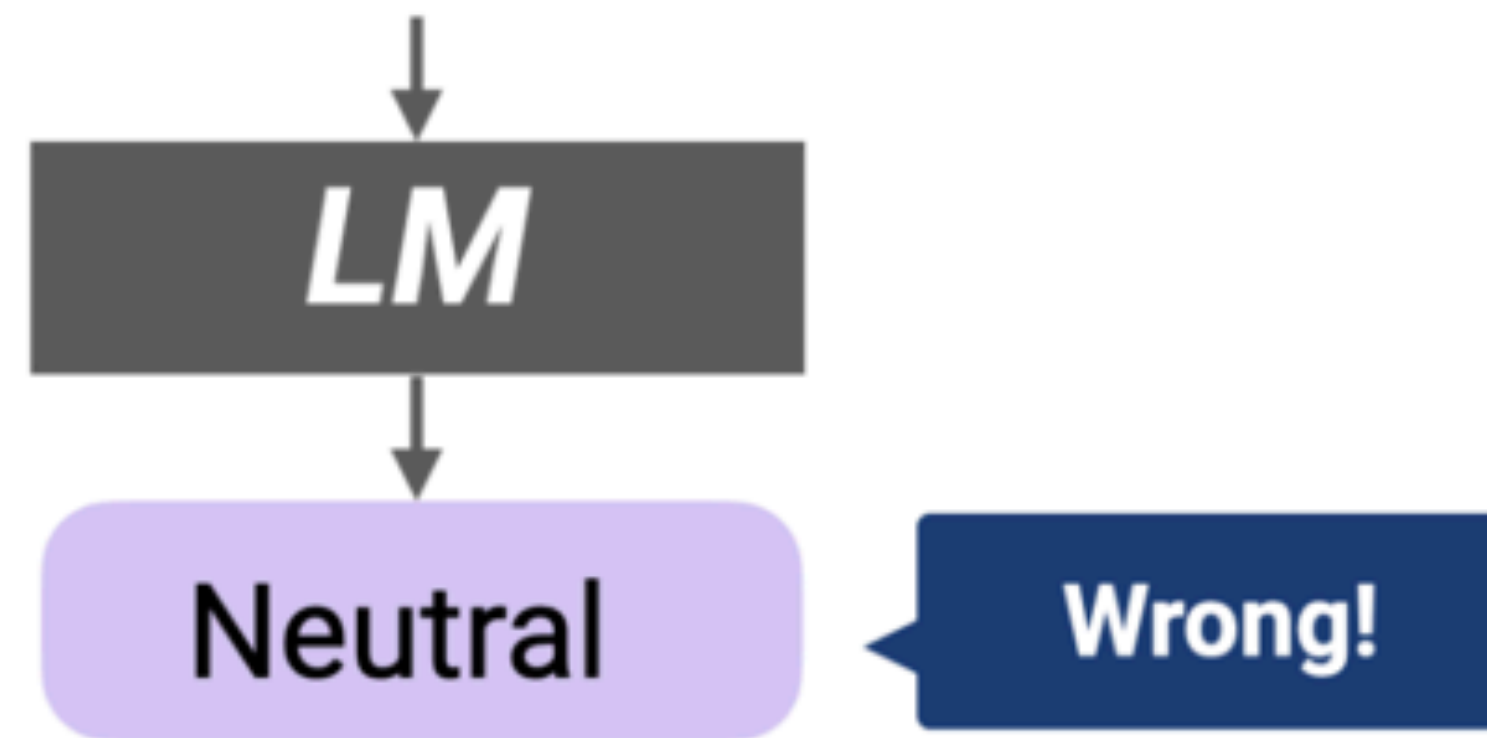
\n Negative

In case you are interested in learning more about ...

\n Positive

The company anticipated its operating profit to improve. \n _____

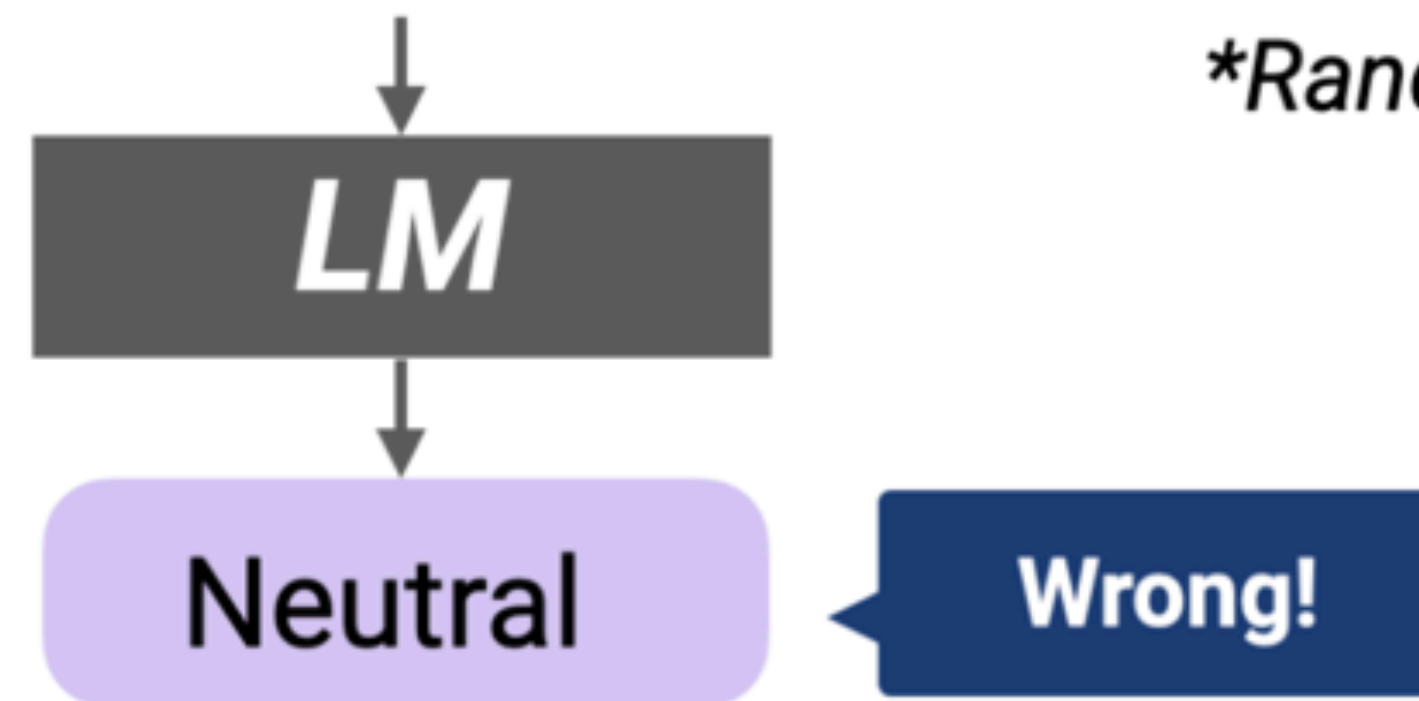
**Randomly Sampled from CC News*



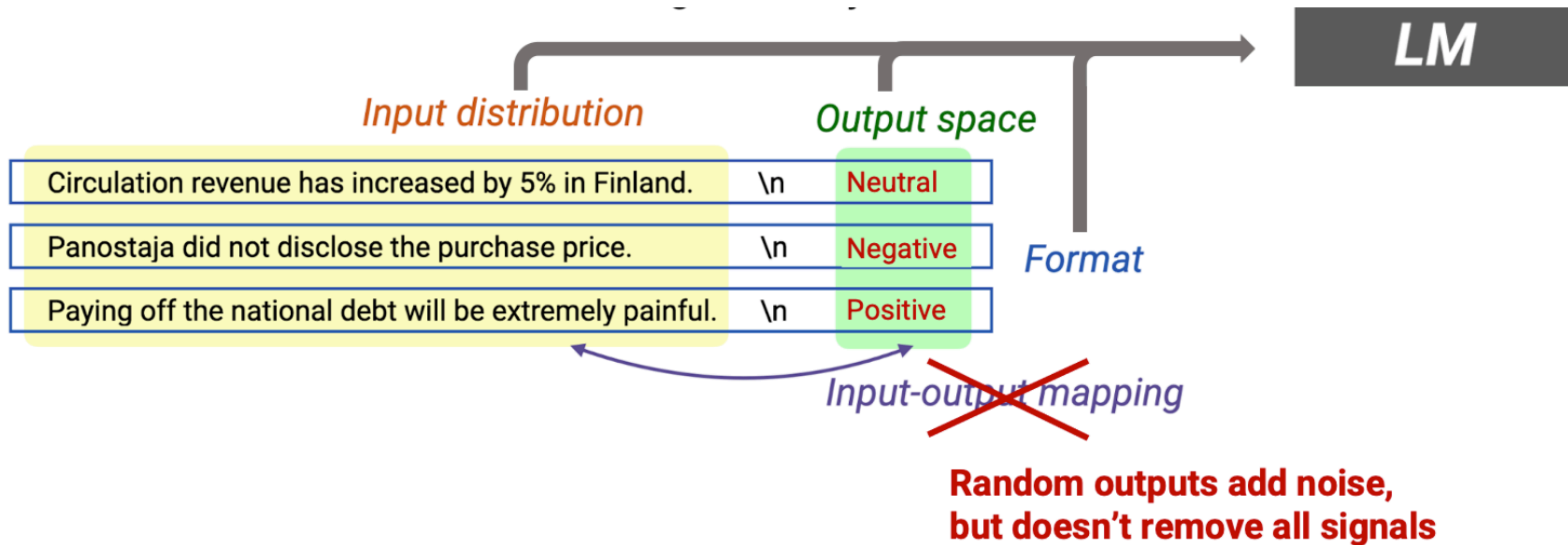
The input distribution matters: using inputs from an out of domain corpus causes a large performance drop

Circulation revenue has increased by 5% in Finland. \n Unanimity
Panostaja did not disclose the purchase price. \n Wave
Paying off the national debt will be extremely painful. \n Guana
The company anticipated its operating profit to improve. \n _____

**Random English unigrams*



The output distribution matters: using labels that are random English unigrams causes a large performance drop



Training examples (truncated)

```
beet: sport  
golf: animal  
horse: plant/vegetable  
corn: sport  
football: animal
```



Test input and predictions

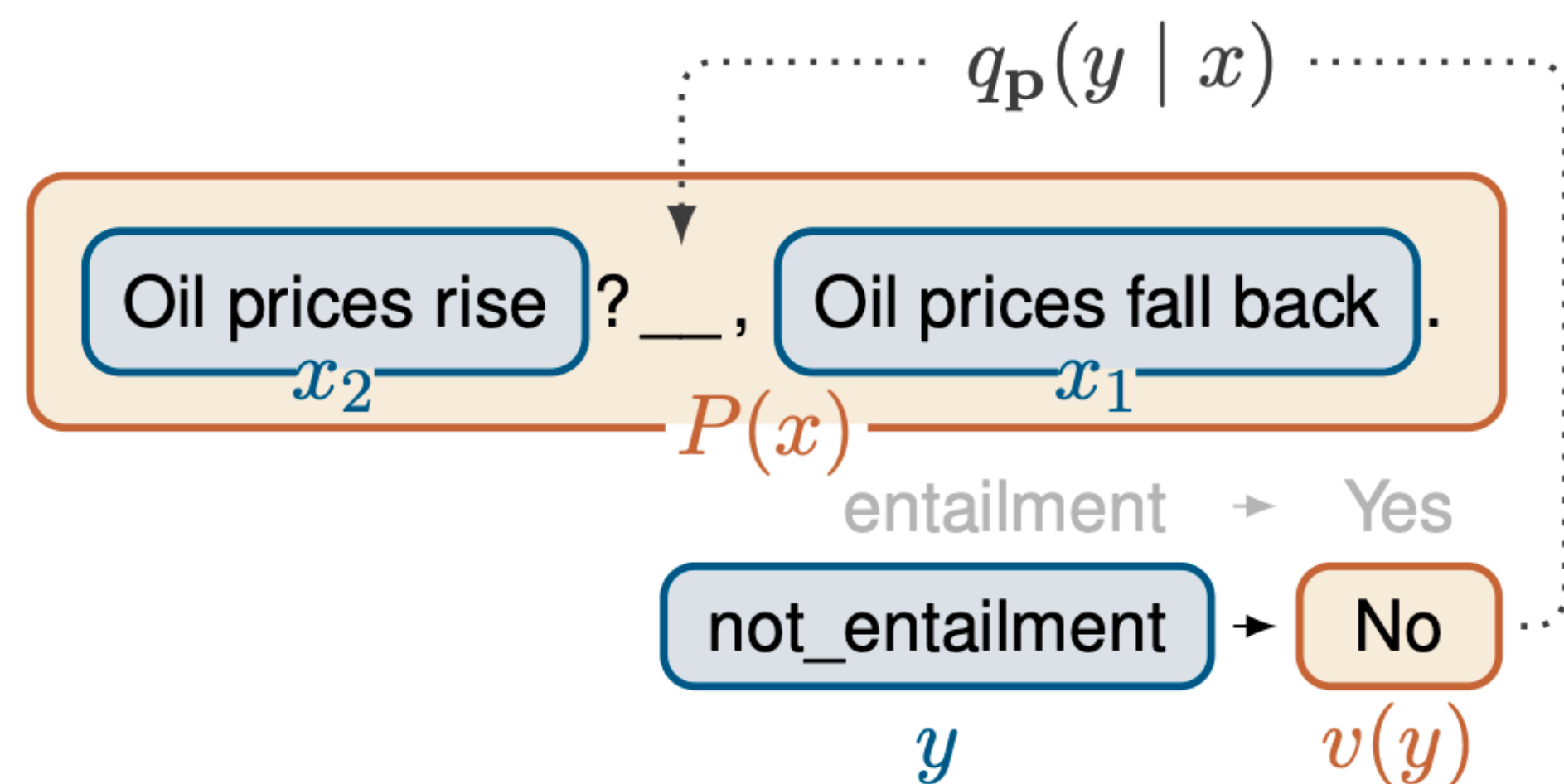
```
monkey: plant/vegetable ✓  
panda: plant/vegetable ✓  
cucumber: sport ✓  
peas: sport ✓  
baseball: animal ✓  
tennis: animal ✓
```

An example synthetic task with unusual semantics that GPT-3 can successfully learn. A modified figure from Rong.

Efficient few-shot learning

Prompt tuning: few-shot with smaller LMs

iPet: better pre-training for each task improves accuracy for small LMs



test	GPT-3	175,000	71.8	prompt
	PET	223	74.0	prompt FT
	iPET	223	75.4	prompt FT
	SotA	11,000	89.3	full FT

Figure 2: Application of a PVP $\mathbf{p} = (P, v)$ for recognizing textual entailment: An input $x = (x_1, x_2)$ is converted into a cloze question $P(x)$; $q_{\mathbf{p}}(y | x)$ for each y is derived from the probability of $v(y)$ being a plausible choice for the masked position.

Summary

Few-shot Prompting

<https://arxiv.org/abs/2102.09690>

- Sensitivity to prompts (Zhao et al 2021):
 - *Majority label bias* — if label distribution is not balanced
 - *Recency bias* — label at the end may be repeated.
 - *Example ordering*